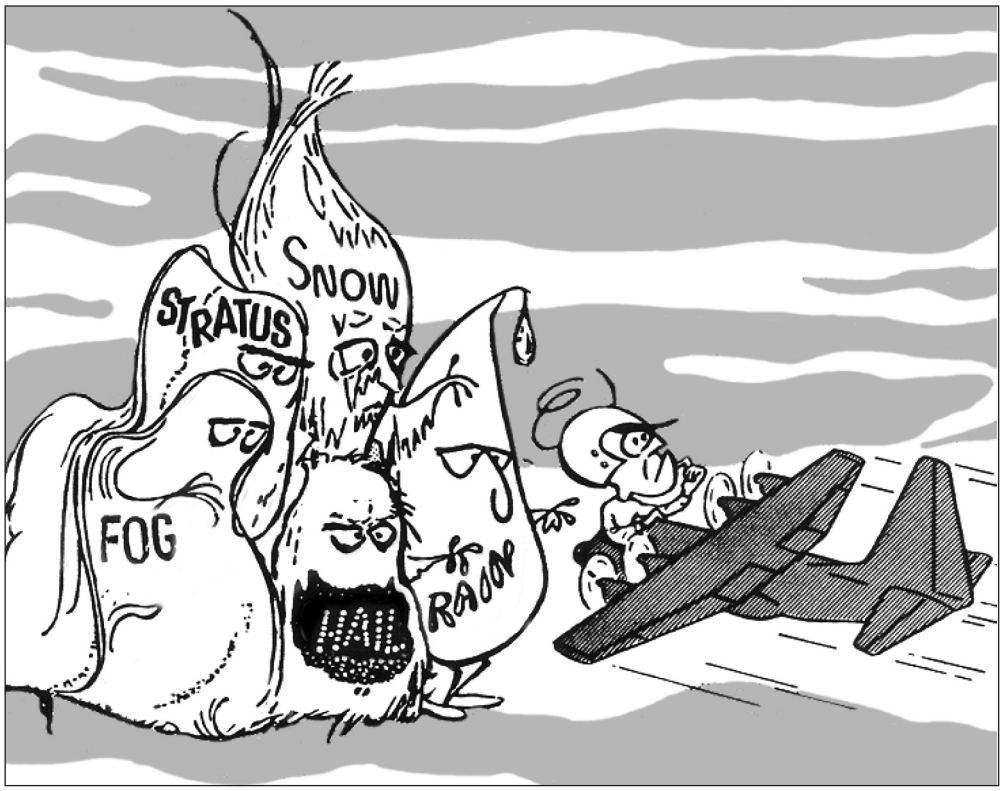
Chapter 1



You are going to have to make decisions regarding the weather many times in your flying career. Since most weather forms as water changes between vapour, liquid and ice, some knowledge of the behaviour of moisture in the atmosphere will help you make the correct decisions.

CHAPTER 1

MOISTURE IN THE ATMOSPHERE

1. Clouds and precipitation are obvious forms of water in the atmosphere, but there is water vapour mixed with the air even on a bright, cloudless day. The amount of the water vapour varies and, although we cannot sense it as easily as air temperature, we can feel the difference in the humidity of a hot, muggy day as compared to that of a cold, crisp one.

2. A very important exchange of energy occurs when water in the atmosphere changes between vapour and cloud which is basic to the development of weather. On the scale of atmospheric processes, the amount of energy in this change is far from trivial. The energy associated with one thunderstorm, for instance, comes mainly from this source and is equivalent to a dozen or so Hiroshima-type bombs. A hurricane releases almost that much energy in a second.

Changes of State

3. The moisture in the atmosphere originates mainly from evaporation from the oceans and lakes and from transpiration from the vegetation of the earth's surface. While the amount remains greatest near these sources it is mixed throughout the lower 20,000 to 40,000 feet of the atmosphere. Since cloud is formed when water vapour changes to liquid water droplets or ice crystals, it is only within this layer that clouds and precipitation occur.

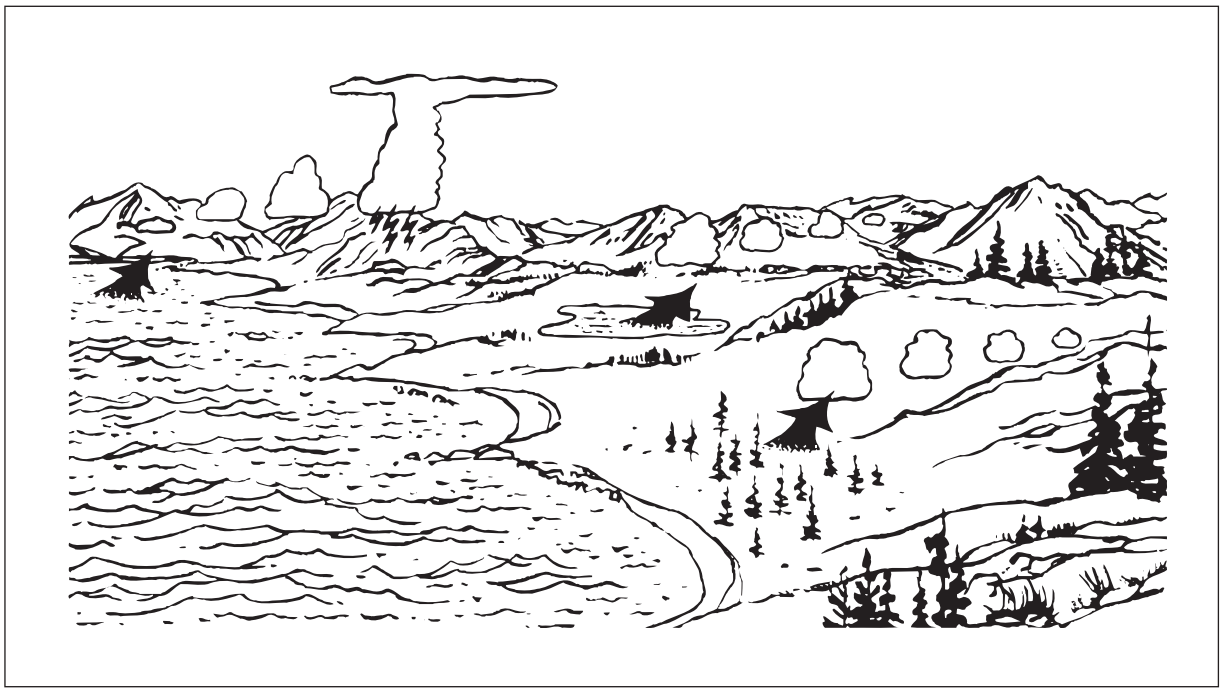


Figure 1-1 Source of atmospheric moisture

4. Water can change back and forth from gas, liquid and ice at ordinary atmospheric pressures and temperatures. As a gas, water is in a high energy state, with its molecules moving freely and rapidly. As a liquid it is in a medium energy state and as ice it is in a low energy state with the molecules moving only slightly. Water as high energy water vapour can condense to lower energy liquid and remain at the same temperature. As it slips into this lower energy state it releases energy to the atmosphere as heat. This heat is called the "Latent Heat of Vaporization." In the reverse process when liquid evaporates to vapour or gas at the same temperature, it absorbs the same amount of energy from the atmosphere. This is what causes cooling of the skin from perspiration. In this case, the heat required to evaporate the perspiration is taken from the skin, thus cooling it.

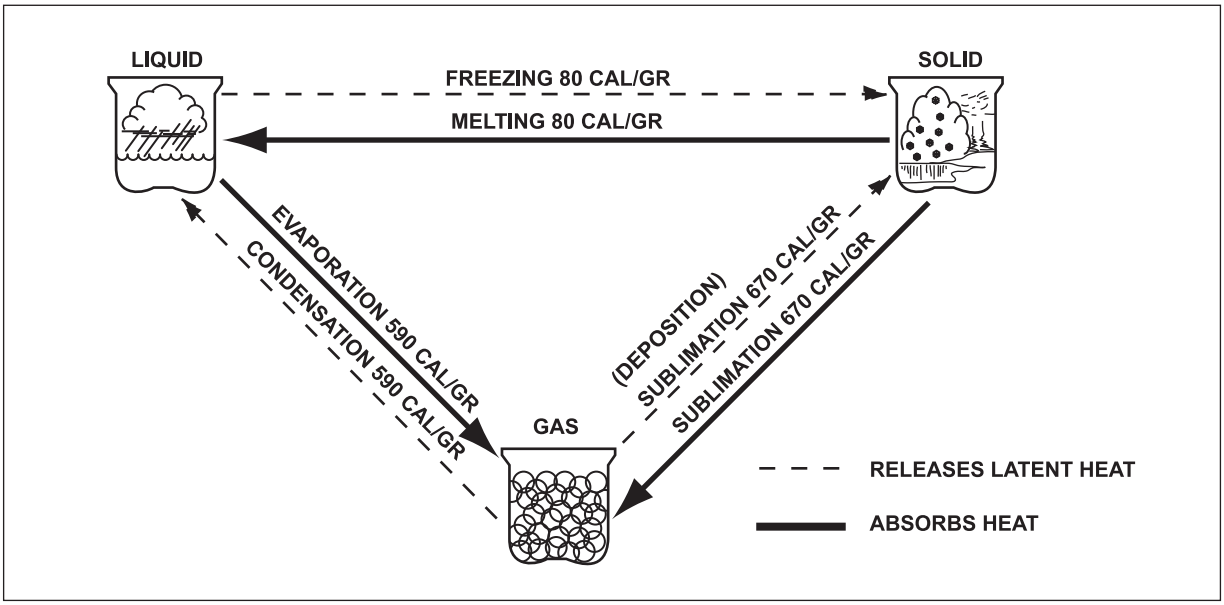


Figure 1-2 Change of state

5. During the change from liquid to ice, heat is released into the atmosphere, and from ice to liquid, heat is absorbed. In these cases the heat gained or lost is called the "Latent Heat of Fusion." Ice can change directly to water vapour and water vapour directly to ice. In both cases the process is called "Sublimation" and the heat gained or lost is called the "Latent Heat of Sublimation." Heat is released in the change to ice and absorbed in the change to vapour and it is equal to the latent heat of vaporization plus the latent heat of fusion. To avoid confusion, the term "deposition" is starting to be used when describing the change of state of gas directly to ice. The energy previously referred to in thunderstorms or hurricanes comes from the release of the various latent heats as water vapour changes to water droplets and then to ice crystals.

Moisture Content

6. There is a limit to the amount of water that can exist as vapour in the air at any given temperature. When this limit is reached, saturation occurs and any cooling will cause condensation and cloud will form. This is illustrated in Figure 1-3, which shows the water vapour content when the air is saturated at various temperatures. From this diagram you can see that at 30°C, 30 grams of water vapour can

exist in a cubic metre of air (point A). At 27°C, only 25 grams per cubic metre can exist (point B), at 6°C, 7 grams (point C) and at 3°C, 6 grams (point D). If the temperature drops 3°C from 30°C to 27°C, 5 grams of water vapour per cubic metre will condense. A similar 3°C drop from 6°C will cause only 1 gram of water vapour per metre to condense. Cooling of saturated air at warmer temperatures causes more water vapour to condense than the same amount of cooling of saturated air at colder temperatures.

7. Relating this fact to latent heat and energy, more energy is released during condensation at warmer temperatures than at cooler temperatures for an equivalent amount of cooling. For this reason, the most violent weather, such as hurricanes, tornadoes, and severe thunderstorms, occurs in very warm, moist air.

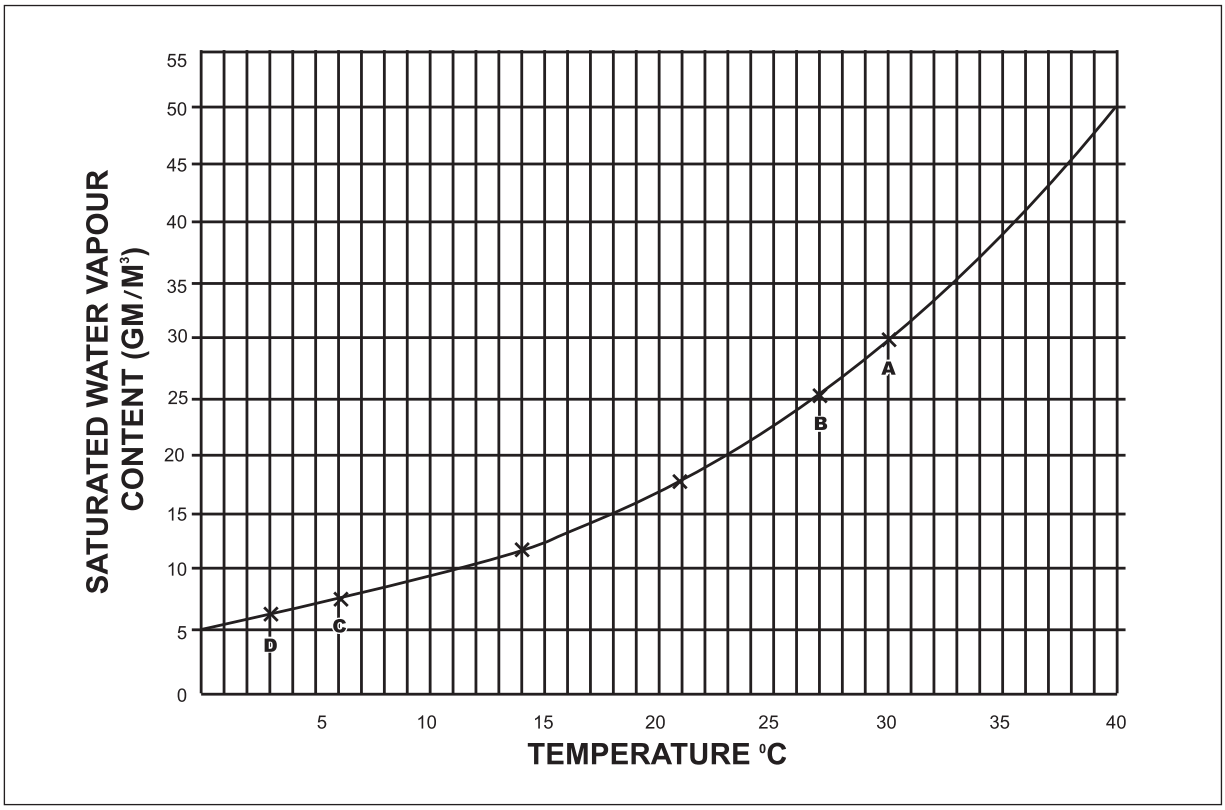


Figure 1-3 Variation of saturated water vapour content with temperature

Dew Point

8. You can assess the amount of water vapour in the atmosphere from the dew point. The "Dew Point" is the temperature at a given pressure to which air must be cooled to cause saturation. The difference between the temperature and the dew point is an indication of how close the air is to saturation. This difference is referred to as the "Spread," and a spread of zero means that the air is saturated. The higher the dew point, the more water vapour there is in the air. Look at Figure 1-3. You can see that at a dew point of 30°C there are 30 grams of water vapour in a cubic metre of air, but at 6°C there are only about 7 grams.

Relative Humidity

9. The "Relative Humidity" compares the amount of water vapour in the air with the amount that it could hold if it were saturated and this is expressed as a percentage. One hundred per cent relative humidity indicates saturation. Look again at Figure 1-3. Air with a 100% relative humidity at 30°C holds 30 grams of water vapour per cubic metre but at 6°C it holds only 7 grams. In both cases, there is 100% relative humidity, but the actual water content is quite different. Relative humidity, therefore, does not give as direct an indication of moisture as does dew point. For this reason, dew point and not relative humidity is provided for most aviation purposes.

Condensation Nuclei

10. In the condensation process, water changes from its gaseous state into tiny water droplets, and in the sublimation process, it changes to tiny ice crystals. A peculiar feature of these processes is that they will not occur even when the air is cooled below the saturation temperature unless small particles called condensation or sublimation nuclei are present in the atmosphere. The nuclei are formed of sea salts from the evaporation of ocean spray or of tiny solid particles formed during combustion in industrial areas or from forest fires. They are always present in sufficient numbers that condensation will occur at 100% relative humidity. If they are present in very large numbers, however, for example in industrial areas or over oceans, they can cause condensation to occur even before the air is completely saturated. This is one reason why there is so much cloud over the oceans and industrialized areas.

Aircraft Performance

11. The density of air decreases with decreased atmospheric pressure, increased temperature and increased water vapour content. Aircraft performance varies with the air density so that at any given pressure the performance is better in denser air. Although temperature is the major controlling factor in this situation, moisture does play a small part so that for some critical aircraft operations, the moisture content of the air must be considered.

Summary - Chapter 1

- Cooling causes saturation; further cooling causes condensation or sublimation and the formation of cloud.
- The amount of water condensed is greater when warmer saturated air is cooled than when colder saturated air is cooled.
- During evaporation, melting and sublimation (ice to vapour), heat is absorbed by the moisture.
- During condensation, freezing and sublimation (vapour to ice), heat is released to the atmosphere.
- The dew point is the temperature to which air must be cooled (at constant pressure) to cause saturation.
- The relative humidity is the amount of water vapour in the air compared with what it would hold if it were saturated.
- Nuclei must be present for condensation or sublimation to occur. If they are present in abundance, these processes can occur at less than 100% relative humidity.
- Aircraft performance decreases slightly in air having a high water vapour content.